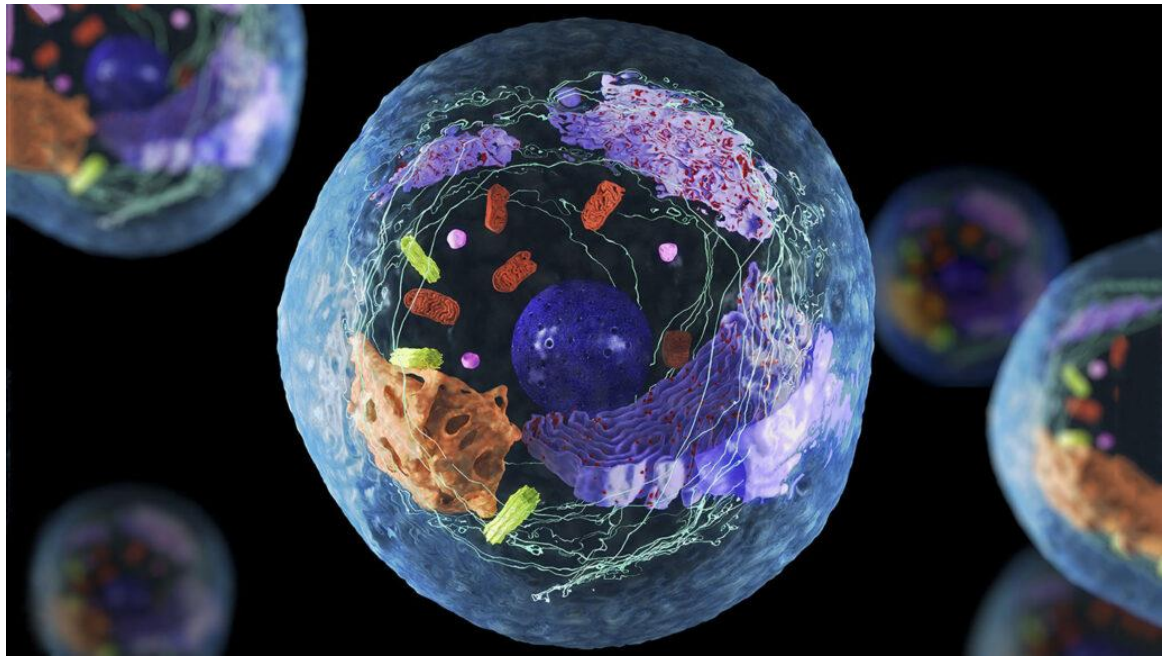
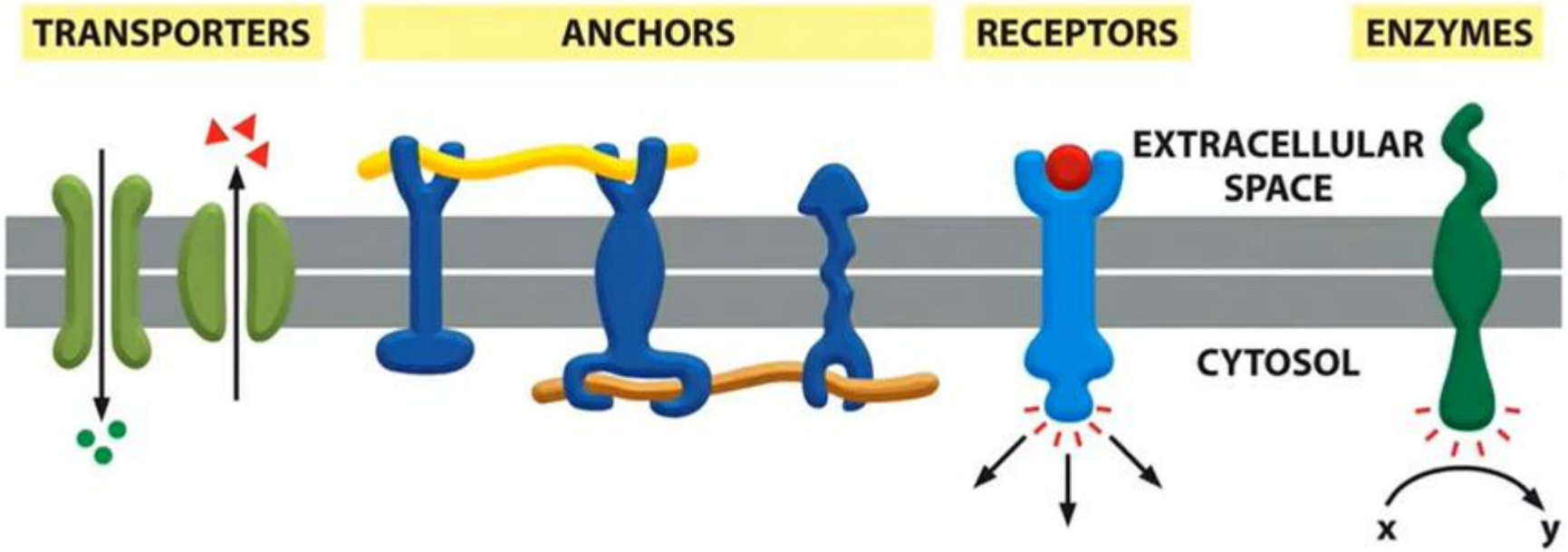
Membrane transport

Lecture 15



Membrane proteins

- The lipids of the membrane are largely involved in structural integrity of cell.
- Lipids are not functional, and hence can't do much activity.
- The activity and function of the cell membrane is carried out by membrane proteins



Most common function of the membrane proteins is transport.

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Membrane transport: three basic mechanisms

- Simple diffusion: movement down a concentration gradient without a protein.
- Facilitated diffusion: protein-assisted movement down an electrochemical gradient.
- Active transport: energy-coupled movement that can occur against a gradient.
- For ions, both concentration and electrical gradients contribute to the driving force.

Types of Cellular Transport

Passive Transport

Cell doesn't use energy

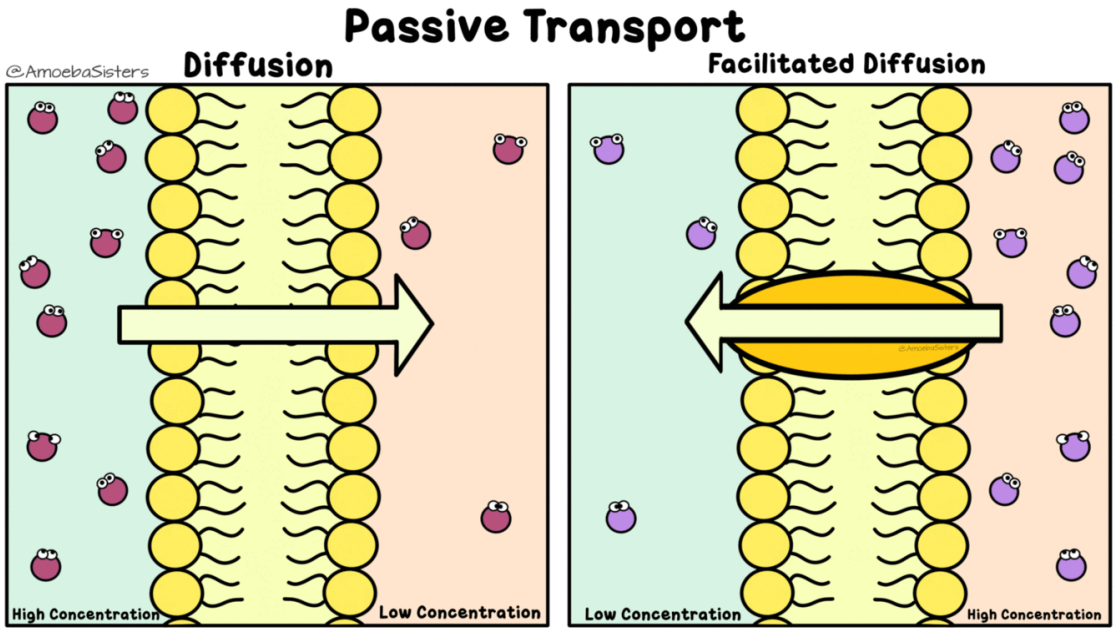
1. Diffusion
2. Facilitated Diffusion
3. Osmosis

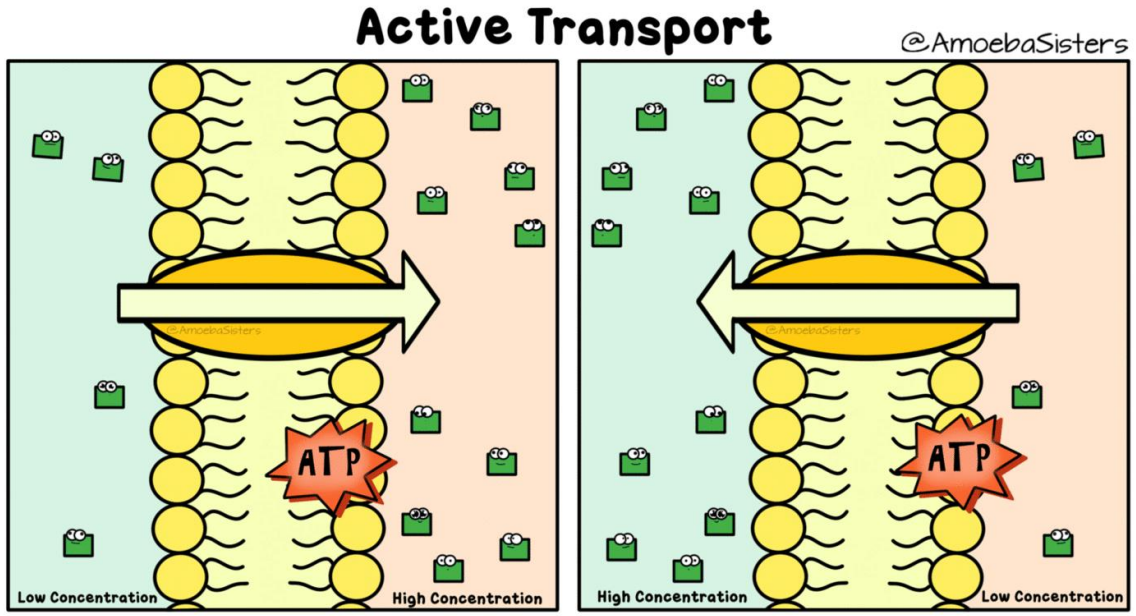
Active Transport

Cell does use energy

1. Protein Pumps
2. Endocytosis
3. Exocytosis

Types of Cellular Transport

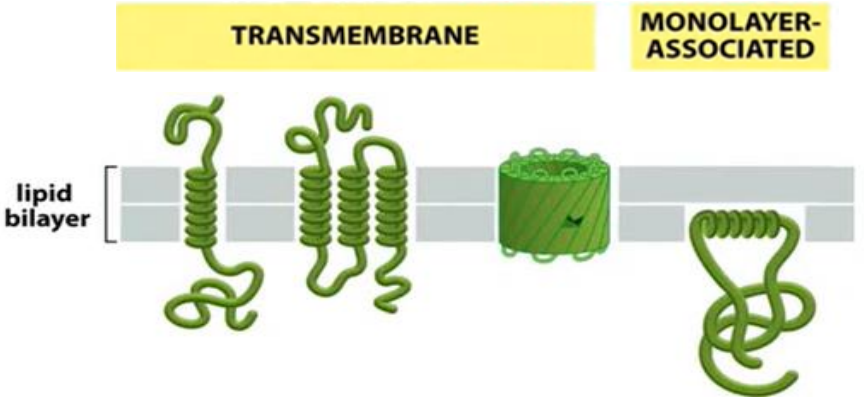




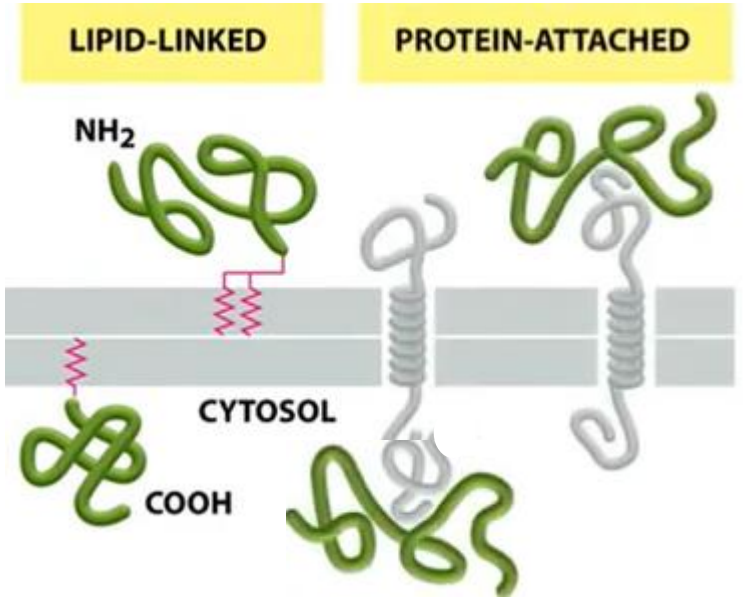
https://www.amoebasisters.com/

Membrane protein's association with bilayer

- Proteins embedded in the bilayer have some or all of their chain as hydrophobic



- Some membrane proteins are not embedded in membrane at all
  - They are either covalently linked to a phospholipid head
  - Or, they are non-covalently associated with already embedded membrane proteins

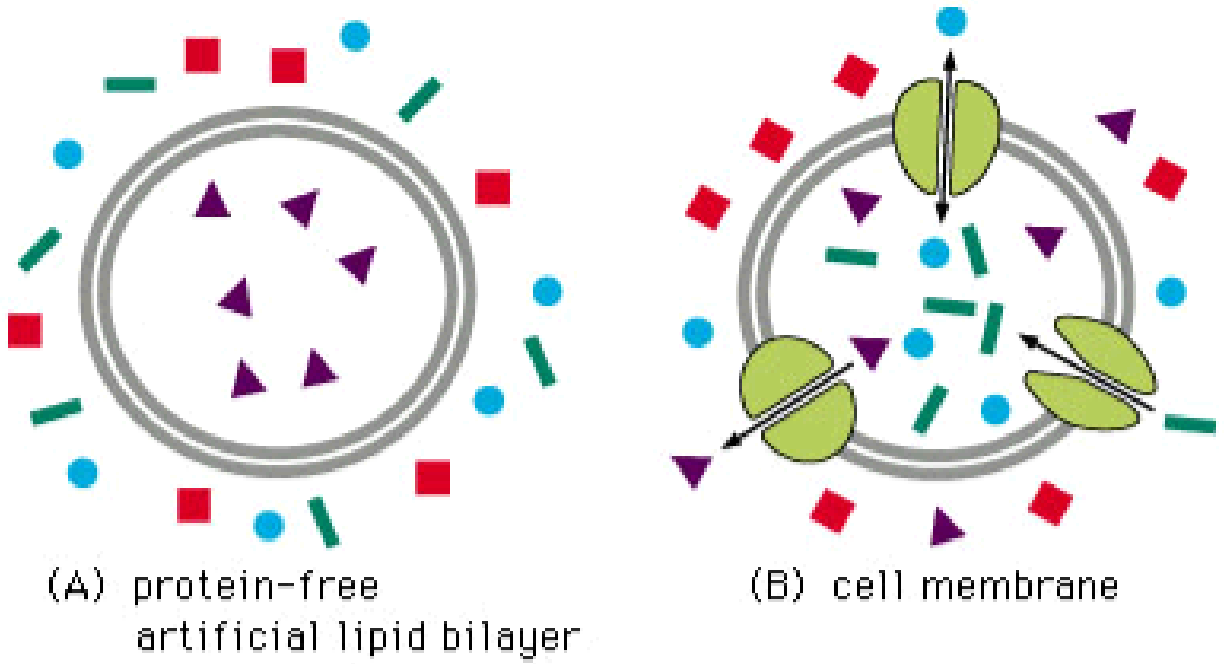


- Embedded proteins are called INTEGRAL and non-embedded proteins are called PERIPHERAL

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Transport proteins – what we have learned?

- Cells survive solely by exchanging molecules with their environment
- Because the membrane is made of hydrophobic material, diffusion of hydrophilic material is not possible at relevant time scale
- To bypass this inher e transport proteins
- Transport proteins p ne cell membrane that allo
- The transport is dor



Ion concentration

- Ion concentration differs inside and outside of the cell
- Most of the membrane transport proteins are engaged in the transport of ions.
- Cells intentionally maintain ionic concentration inside that is different than the concentration outside
- Cells regulate the movement of sodium (Na^+), potassium (K^+), chlorine (Cl^-) and proton ions across the membrane.
- There movement is critical for cellular survival.

Compartmentalization maintains selectivity

- Sodium is more concentrated outside the cell, while potassium is in higher concentration inside the cell.

- If the charge balance were too skewed, the electrical/ionic forces would overwhelm the structural integrity of the cells and tear them apart.

TABLE 12–1 A COMPARISON OF ION CONCENTRATIONS INSIDE AND OUTSIDE A TYPICAL MAMMALIAN CELL

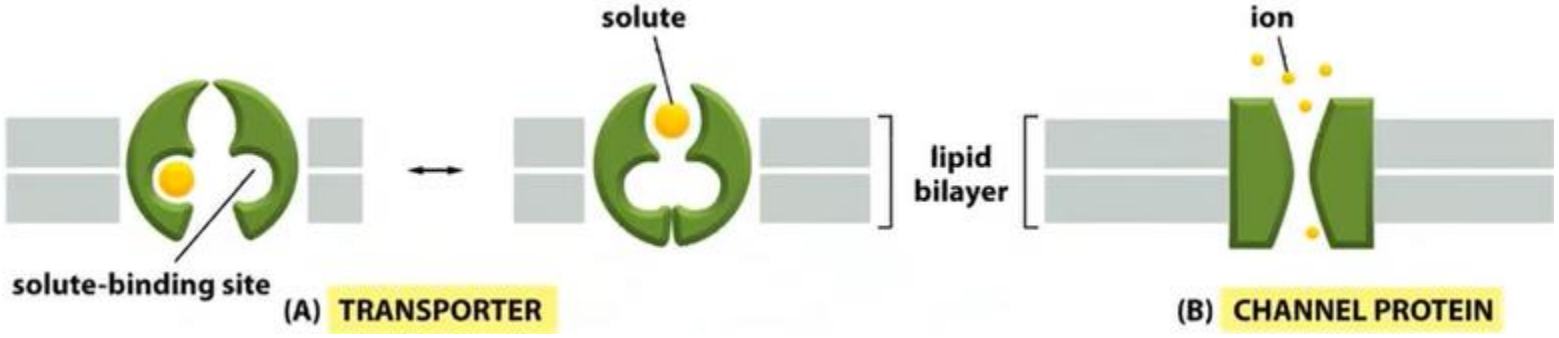
COMPONENT	INTRACELLULAR CONCENTRATION (mM)	EXTRACELLULAR CONCENTRATION (mM)
Cations		
Na^+	5–15	145
K^+	140	5
Mg^{2+}	0.5	1–2
Ca^{2+}	10^{-4}	1–2
H^+	7×10^{-5} ($10^{-7.2}$ M or pH 7.2)	4×10^{-5} ($10^{-7.4}$ M or pH 7.4)
Anions*		
Cl^-	5–15	110

* The cell must contain equal quantities of positive and negative charges (that is, be electrically neutral). Thus, in addition to Cl^-, the cell contains many other anions not listed in this table; in fact, most cellular constituents are negatively charged (HCO_3^-, PO_4^{3-}, proteins, nucleic acids, metabolites carrying phosphate and carboxyl groups, etc.). The concentrations of Ca^{2+} and Mg^{2+} given are for the free ions. There is a total of about 20 mM Mg^{2+} and 1–2 mM Ca^{2+} in cells, but this is mostly bound to proteins and other substances and, for Ca^{2+}, stored within various organelles.

Transport proteins

- TRANSPORTER or pump is a protein that contains moving parts and can transfer from one side of the membrane to the other
    - Transporters can usually move molecules from in to out or out to in; but they do not move simultaneously in both directions
    - They often move things against their concentration gradients

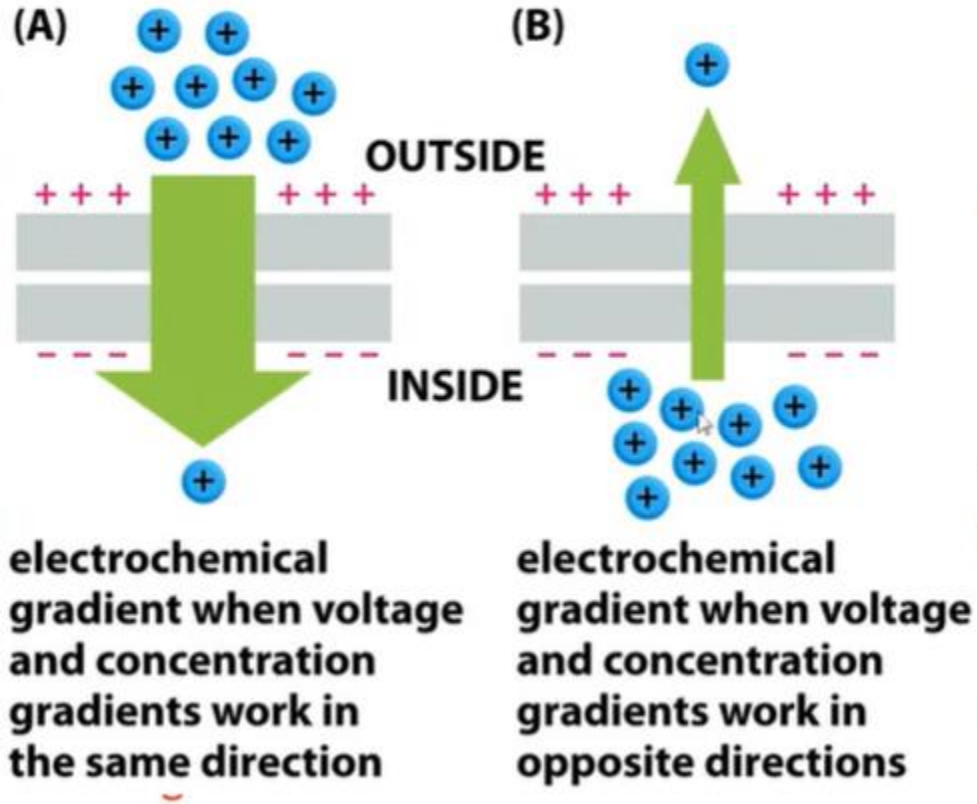
- CHANNEL is a protein with no moving parts
    - They are usually like pores
    - They create small hydrophilic holes in the membrane which can only be used by a select few molecules.
    - Most common are ION CHANNELS as they are specifically for small inorganic ions.
    - Molecules move across these channels by passive transport



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Concentration gradients and electrical forces drive transport

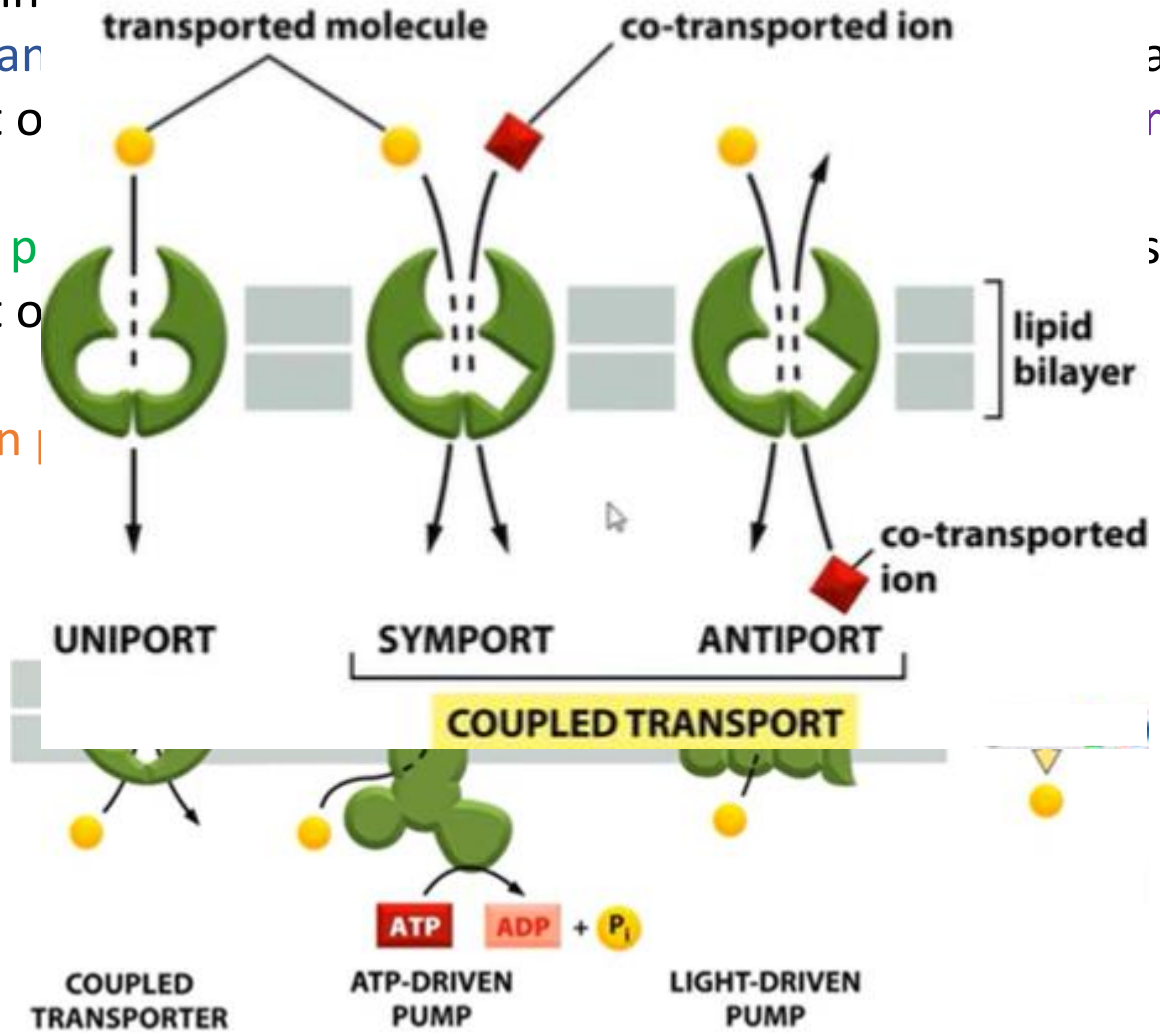
- Some transporters, like the glucose transporter, will move molecules in either direction
  - After a meal, glucose is higher in the blood than in the cell, so it moves into the cell
  - After fasting, glucose is higher in the cell than in the blood. So it moves out of the cell
  - The direction of m
- It's not that straightforv
  - In localized regions of charge inside and outside the cell
  - It means there is a co t as well as ionic attraction.
  - Inside the cell is usual pelling those that are negative



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Membrane also moves molecules against their electrochemical gradient

- Sometimes, cells need to work against the electrochemical gradient to move molecules.
- This can be achieved in [illegible] different [illegible]
  1. Coupled tran[illegible] adient to power the movement o[illegible] rt
  2. ATP driven p[illegible] sis to force the movement o[illegible]
  3. Light driven [illegible]



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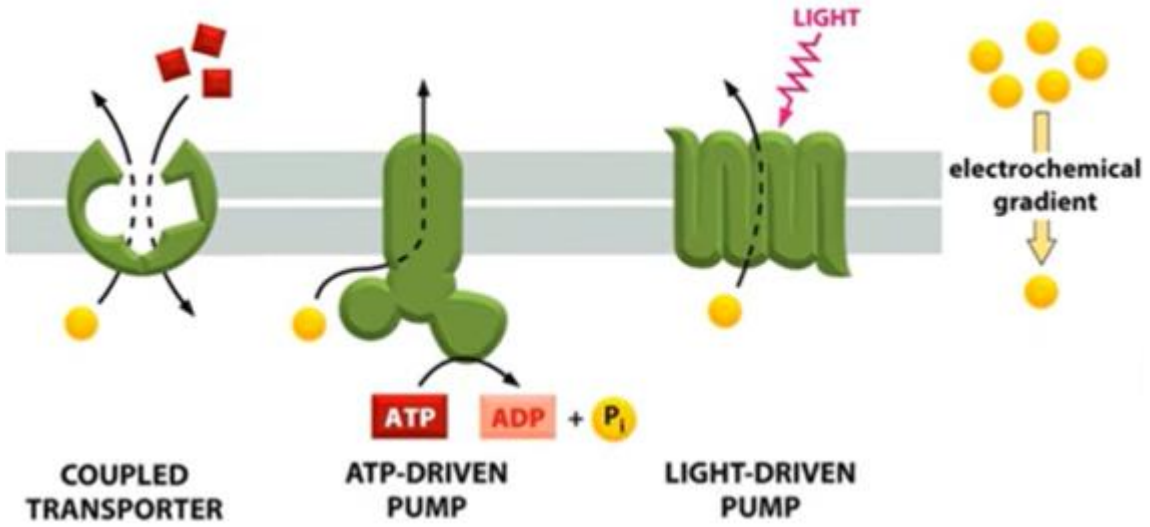
Membrane also moves molecules against their electrochemical gradient

- Sometimes, cells need to work against the electrochemical gradient to move molecules.

- This can be achieved in three different ways:
    1. Coupled transporters: they use one molecule moving with its gradient to power the movement of another against its gradient – symport and antiport

    2. ATP driven pumps: these use energy obtained from ATP hydrolysis to force the movement of molecules

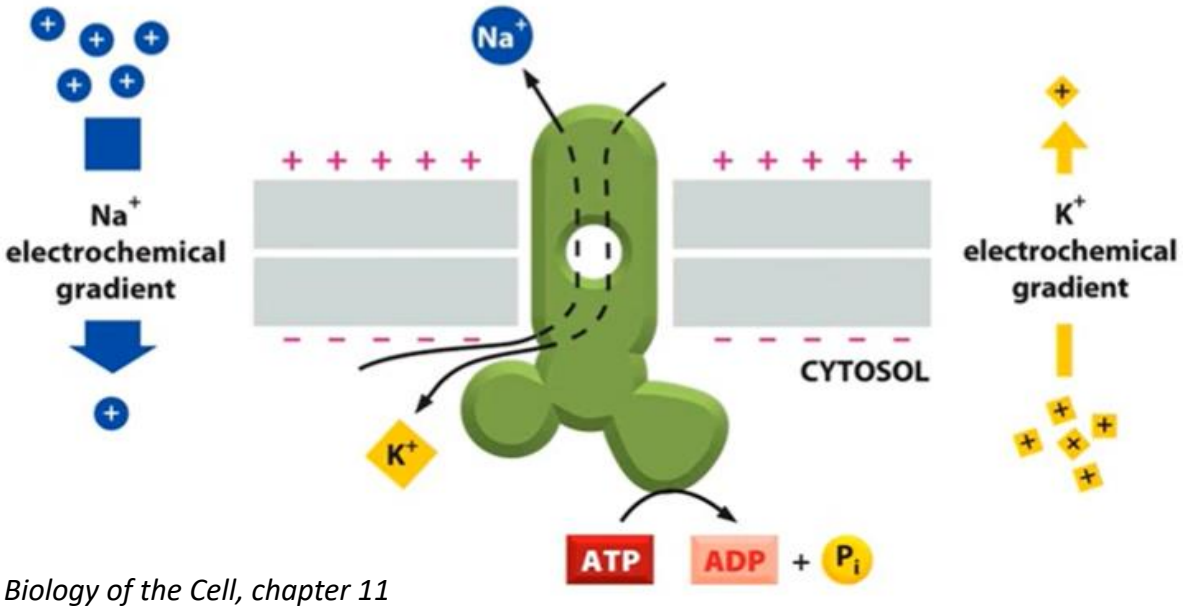
    3. Light driven pumps – uses the energy of light to molecules

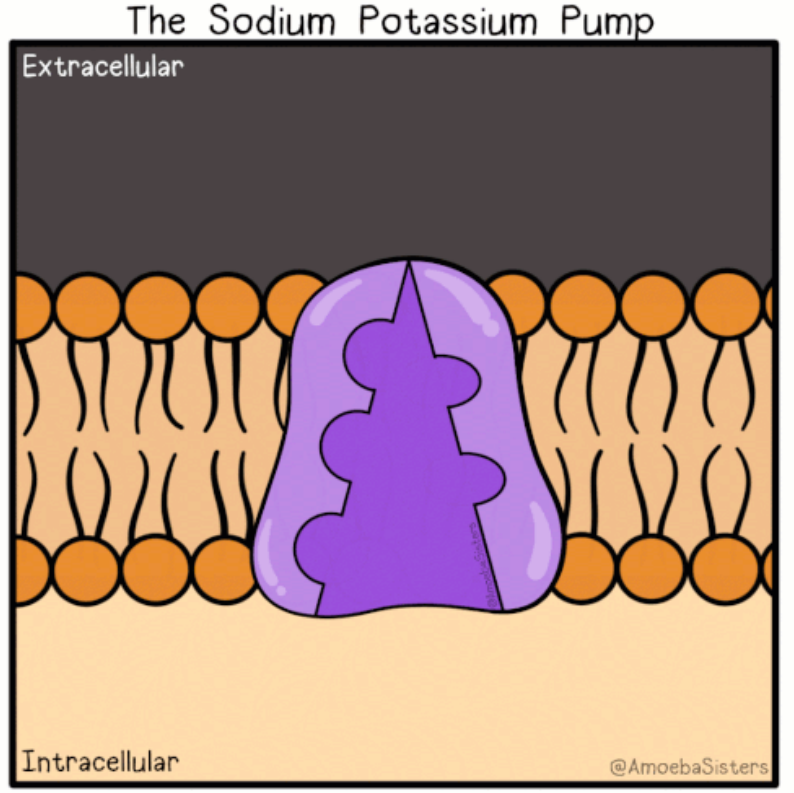


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The Sodium-potassium pump

- Sodium pump has dual role:
  - It pumps sodium out of cell, against its gradient
  - It also allows potassium into the cell
- This pump is ATP dependent
- This pump alone is responsible for 30% of cell's ATP consumption
- The pump manages to keep the intracellular concentration of sodium ~10-30 times lower than extracellular concentration





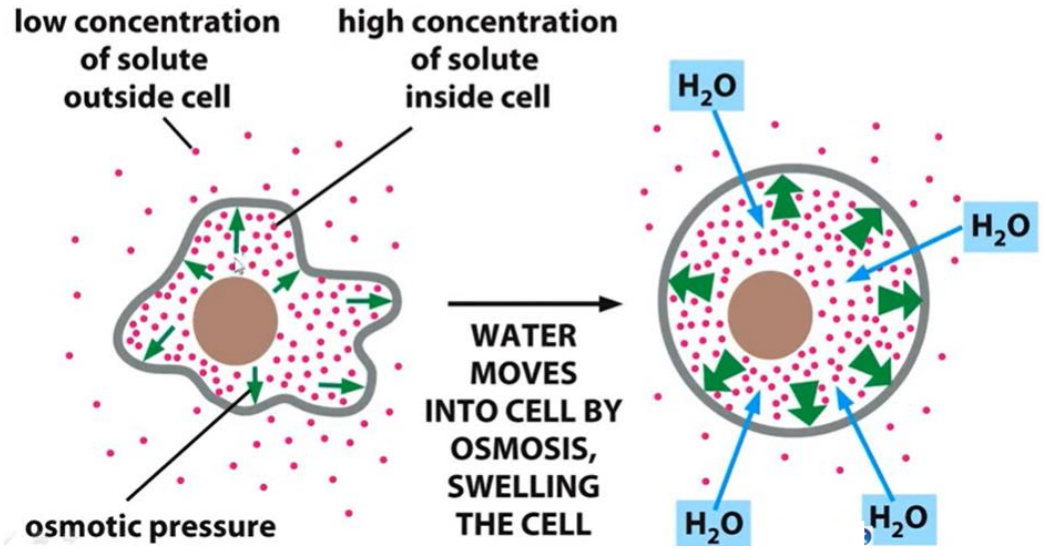
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The Sodium-potassium pump maintains osmotic balance

- Water will diffuse from where there is more water to where there is less – OSMOSIS

- For this reason, as sodium levels climb in the cell, water level also rises
    - Both diffusing through the membrane directly and flowing in through specific water channels called AQUAPORINS

- This puts the cell at risk of bursting
    - By pumping Sodium back out again, the pump also helps to keep water from staying in the cell



The membrane Potential

- When we deal with charged matter, we are dealing with electricity
    - When charged things move, current is generated and is a form of kinetic energy
    - When charged things can't move, but want to, a charged potential is generated and measured in volts

- At the cell membrane, the inside of the cell is negatively charged while outside is positively charged
    - Opposite attract and these ions want to move
    - But the membrane does not allow them to move, creating a potential
    - This potential is known as Membrane Potential and every cell has one

- The resting membrane potential of most cells is 20 to 200 mV

The membrane Potential

- The forces largely balance
    - The electric gradient is opposed, but balanced by the different gradients of individual ions
    - Sodium has an imbalance of force
    - Any flow of ions across the membrane creates current and changes the membrane potential, making it more positive

- Since ion channels are highly selective, and only let specific ion through, to accommodate this change in membrane potential and eventually restore it back to rest, other ion channels must open as well

- This implies that opening of ion channels are regulated to change and restore membrane potential
    - Ion channels are gated and many things can control the gate